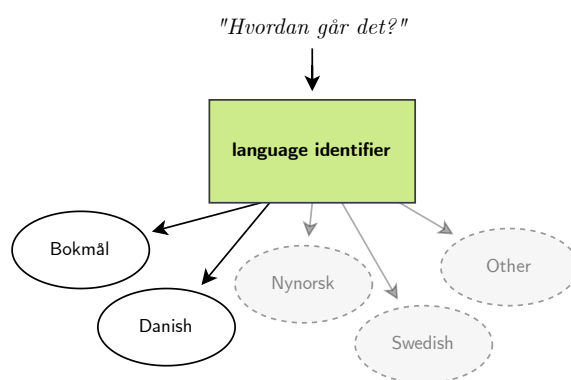


IN5550, 2024 Home Exam

Scandinavian Language Identification

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Introduction

Compared to the complex tasks you have successfully tackled in this course (such as summarization), language identification might seem like a piece of cake. Indeed, it is regarded to be a solved task by many NLP researchers – those that most likely do not speak any non-major language. Because language identifiers are at best terrible and most likely non-existing for such languages, which is a major hurdle for developing NLP tools for them.

Language identification is at the heart of preprocessing for any large-scale text corpus. If you train a Swedish language model, you will train it on texts that are identified as ‘*Swedish*’. If you train a Nynorsk machine translation model, it will learn to produce text that is identified as ‘*Nynorsk*’. The performance of these large and complex models is thus limited by the accuracy of a simple language identification tool used to create the training corpus.

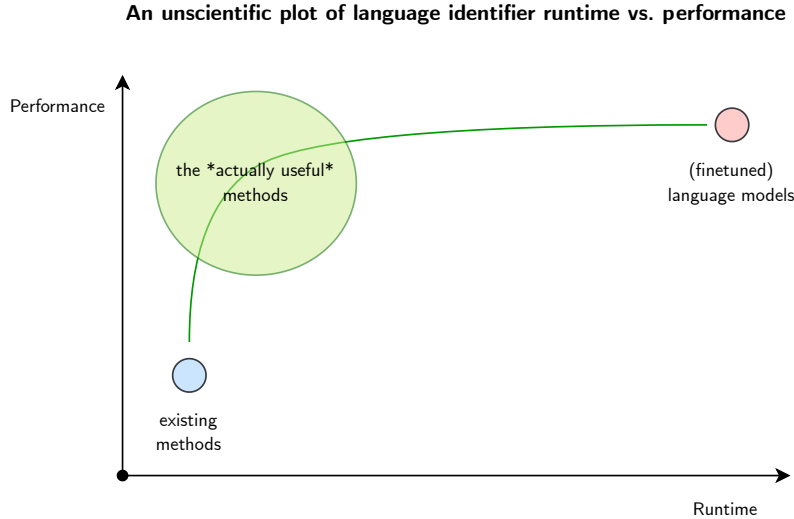


Figure 1: An illustrative plot that shows the hypothesized pareto front of language identification performance at different model complexities.

The existing identifiers are relatively poor when it comes to distinguishing between the Scandinavian languages, your task is to solve this problem :) A tempting solution would be to simply finetune a large language model on this task – but remember that the primary use of this tool is to filter large quantities of text, therefore, you should be very mindful about the computational complexity of these models, as illustrated in Figure 1. The proposed direction for this project is to use finetuned language models as a means to improve the training of a cheaper method (but you are welcome to diverge from this direction!)

We focus on the Scandinavian languages Danish, Norwegian and Swedish, as they are closely related and pose an interesting challenge when it comes to language identification. Some language identification systems treat Norwegian Bokmål and Norwegian Nynorsk as the same, while some distinguish them. We will distinguish between them in our setup. We do not include other languages spoken in Scandinavia such as Sami languages or Elfdalian, but include a label *other* that is used for any language that is not Swedish, Danish, Bokmål or Nynorsk.

For general instructions regarding the home exam, see the information here:

<https://www.uio.no/studier/emner/matnat/ifi/IN5550/v24/final-exam/>

Data

Gold training dataset

We take the gold training/validation data from Universal Dependencies treebanks (UD; [Nivre et al., 2017](#)).¹ This dataset is clean but not disambiguated (a sentence labeled as Nynorsk is almost guaranteed to be in Nynorsk, but it could also be a valid Bokmål sentence).

Silver training dataset

Neural models like to take shortcuts whenever we allow them to; that can be very bad in our case – it is easy to imagine that there is a lot of texts about skiing in the Norwegian datasets and about ice hockey in the Swedish dataset, then the model might learn to associate any sentence that involves skiing with Norwegian. We would like to decouple the semantics from the language and to do this, we have machine translated all data to create a silver training set. This ensures that the dataset is balanced, both in terms of the number of sentences and in terms of domains covered by each language. The downside is that machine translation can introduce some unwanted artifacts.

Manually filtered evaluation dataset

The evaluation dataset is taken from the test splits of the UD treebanks for each of the included languages, but with additional labels added in cases where sentences could belong to multiple languages. For example, as seen in the illustration above, a sentence like “Hvordan går det?” could be both Bokmål and Danish, but not Nynorsk or Swedish.

Your own data

You do not have to rely on these suggested datasets, we encourage you to look for more data! The UD datasets are mostly sourced from news articles; thus it is definitely a good idea to train your models on data with a broader domain coverage. There are many possibilities to get clean monolingual data, such as the [Wikipedia dumps](#), [Tatoeba](#) or [Open LID dataset](#).

¹ Specifically, we use these UD treebanks: [no_bokmaal](#), [da_ddt](#), [no_nynorsk](#) and [sv_talbanken](#). The *other* languages are randomly sampled from treebanks for other languages.

Language	Gold train	Gold valid	Silver train	Evaluation set
Bokmål	15 435	2 392	22 860	1 939
Danish	4 376	563	34 173	565
Nynorsk	14 029	1 859	24 382	1 511
Swedish	4 278	503	34 252	1 219
Other	8 805	1 165	—	1 742

Table 1: Number of sentences across the datasets and languages. Every sentence in the training and validation splits is annotated with a single label, while the sentence in the evaluation set can be multi-labeled.

Evaluation metric

We provide a convenient script for evaluating your models, it outputs the total runtime and different classification statistics (confusion matrix, accuracy, F_1 score, Mathews’ correlation coefficient, ...).

The script calculates the classification metrics in two ways: **loose** and **strict**. The loose track is meant for models that can only output one language label: whenever the predicted label is among the gold labels, it is considered correct. The strict (hardcore) track is for models that can output multiple labels: the set of predicted labels has to exactly match the gold labels to be considered correct.

You are free to change the script according to your needs, the current version expects you to implement a function (or functor) that takes a sentence (string) and outputs a list of language labels (list of strings).

Baseline models and related work

Language identification is not exactly a hot topic in today’s NLP, a nice recent paper on it is written by [Burchell et al. \(2023\)](#). This paper proposes **the OpenLID model**, which will be our first baseline. This model is a simple linear classifier trained on top of a FastText model, a common technique also used by our second and third baselines: **the original FastText** and the updated **NLLB language identifier** ([Joulin et al., 2017](#); [Team et al., 2022](#)). Our last baseline, **CLD3**, is based on a simple BoW model.

Summary – what you should do

The only required thing to do is to establish the reference performance illustrated in [Figure 1](#): evaluate the existing methods (at least the four suggested ones) and finetune a language model that reaches a good performance on the loose task. This will be the foundation, then it is up to you what to build on top of it.

Some suggested directions (you can pick one of them, but we would be very happy to see a completely different take on this task):

1. Use the slow finetuned LLM to improve performance of a fast model. There are multiple different ways to do it, for example by doing knowledge distillation or by iteratively training a model on data that it misclassifies (according to the slow model).
2. Find a way to reliably recognize ambiguous sentences that are in multiple languages. Then show (hopefully) that your system is better on the strict track.
3. Empirically (dis)prove the hypothesized pareto front from [Figure 1](#), what is the actual relation between runtime and achievable performance?
4. Investigate different way(s) of reaching the *actually useful* area from [Figure 1](#), is deep learning a way to go when we really care about the runtime? Is there a way to use the deep learning method only on the difficult examples and save time? Can the task be solved with very small and fast language models?
5. Analyze the effect of training data on performance. Does the silver dataset help? Is it better to include more sources of data?
6. Use a large generative language model to create a synthetic dataset that works better than any of the existing data sources. Or use such a model to filter out high-quality monolingual corpora for training.
7. Examine cases where your models fail to classify correctly. Which cases are the most difficult? Are these obvious to a human? A dedicated error analysis section can shed important light on how and why the models perform as they do.
8. ...

Questions and answers

Can we use a different language cluster?

Yes, it is possible to work on a different set of languages, especially if you are not comfortable with analyzing the Scandinavian languages. However, the language should be similarly close and you will have to create the training and evaluation datasets yourselves.

We like the topic, but we are more interested in a different take on it – for example, we would like to distinguish different Norwegian dialects, is it possible to work on that?

In general, yes. But you should first consult the topic with the area chairs if it's too different from our suggestions.

We invented a very clever method to do language identification but the results are terrible, is it a problem?

No, it's completely fine to present a negative result in your paper. But then the paper should probably go deeper into analyzing your method and its failure cases – so that the readers can still learn something useful from it.

We invented a very clever method that does not use any deep learning, is it a problem?

No, that's actually great! Assuming it's faster than neural models :) But you will still have to convince the skeptical readers that your method is really better than deep learning alternatives.

I understand why we want fast models, but I'm more interested in making the slow (LM-based) methods really good and robust, can I do that in this task?

Yes, that's definitely possible!

References

- Laurie Burchell, Alexandra Birch, Nikolay Bogoychev, and Kenneth Heafield. 2023. [An open dataset and model for language identification](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 865–879, Toronto, Canada. Association for Computational Linguistics.
- Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. [Bag of tricks for efficient text classification](#). In *Proceedings of the 15th Conference of the European*

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